

Gowtham Ayna Raveendran

Summary

Senior level Data Scientist specializing in **Anomaly Detection** and **Natural Language Processing** with over 10 years of industry experience. Skilled in deploying machine learning models using Spark and building post-deployment observability pipelines and doing **root cause analysis** while also adept at pioneering new solutions for complex challenges. Proficient in **SQL**, **Python** and **R** and cloud technologies (AWS) with expertise in building **Tableau** and **Looker** dashboards.

Professional Experience

- Sep'21 – Present **Data Scientist II - Traffic Protection, Yahoo**
- Architected an end-to-end **ML bot detection pipeline** (XGBoost, Optuna) managed via a highly **reproducible MLOps framework** (MLflow, Hydra, Make). Leveraged **SHAP interpretability** to prove the model successfully identified stealthy scrapers bypassing legacy threat feeds, driving the transition to a structural, behavior-based security posture.
 - Prototyped a denylist-based throttling solution to restrict high-bid-volume CTV devices, estimated **250k/year reduction in operating costs without impacting ad spend**; surfaced a flawed upstream assumption during validation, preventing misaligned production rollout.
 - Engineered and deployed a **model interpretability and observability pipeline** for Real-time scoring model to surface prediction drivers and to detect model drift, establishing proactive model health monitoring and reducing root cause analysis time by **75%**
- Jan '19 – Sep'21 **Data Scientist, Verizon Media**
- Implemented a **CNN-LSTM-CRF Deep learning model** and PySpark pipeline for automated retraining and deployment of **Named Entity Recognition** system, processing over **100M records/day**
 - Developed **Brand Churn Detection model** using Recency-Frequency-Monetary value based segmentation to create targeted advertisement segments generating **\$1M** revenue under influence. This work is **patented**.
 - Fine-tuned a BERT model for targeted sentiment analysis to brand sentiment from user comments, enhancing predictions of downstream Ad Targeting and Churn Detection systems.
- Jan '18 – Jan'19 **Data Scientist, Oath Holdings, Inc**
- Built a hierarchical **product category classifier** using multi-level logistic regression to map items to Google's product taxonomy; leveraged NER-derived features (brand, product line) and optimized training with **Vowpal Wabbit**, improving scalability and model efficiency.
 - Developed a user-brand affinity framework by constructing proxy labels from behavioral signals, generating affinity scores to capture consumer preference beyond purchase data. **Defensive Publication, Feb '19**
- Jun '17 – Jan '18 **Data Analyst, TrueScripts Management Services**
- Built multiple reporting dashboard with **R Shiny** and **Tableau** reducing turn around time by **75%**
 - Engineered and Automated drug repricing system using **R** for estimating potential savings for new clients

Patents and Publications

- Feb'23 **Method and system for user group determination, churn identification and content selection, US20230043820A1, Granted**

Education

- May '17 **M.S. in Data Science, Indiana University Bloomington, GPA:3.84/4.0**
Jun '12 **B.E. in Electronics & Communication, Anna University, India, GPA:8.36/10**

Skills

- Machine Learning Keras, Scikit-learn, Decision Trees, PCA, Ensemble learning - Random Forests, XGBoost, Autoencoders, One class SVMs, Isolation Forests, DBSCAN, K-Means, mlflow, glum
- Tools & Technologies Airflow, Oozie, MLflow, Docker, Git, Looker, Otel, Grafana Tableau, Terraform, Spark, Hadoop, Hive, Vertica, Presto, Athena, Amazon Bedrock Agentcore, SageMaker, EMR, Athena, BigQuery(GCP)
- Languages Python, SQL, R, Bash, Java, Pig, Scala